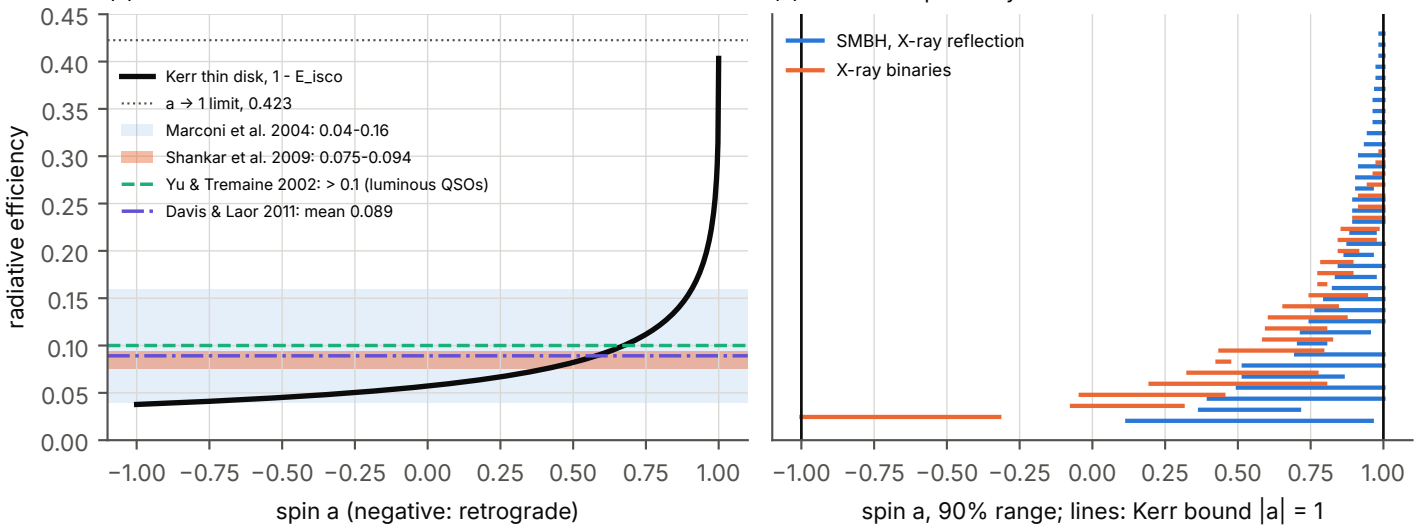
(a) Soltan efficiencies on the Kerr curve
(b) measured spins (Reynolds 2021)
radiative efficiency
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Kerr thin disk, 1 - E_isco
a → 1 limit, 0.423
Marconi et al. 2004: 0.04-0.16
Shankar et al. 2009: 0.075-0.094
Yu & Tremaine 2002: > 0.1 (luminous QSOs)
Davis & Laor 2011: mean 0.089
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
spin a (negative: retrograde)
SMBH, X-ray reflection
X-ray binaries
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
spin a, 90% range; lines: Kerr bound |a| = 1